

Lecture 29 : Public-key cryptography

(Part I)

Introduction

In today's interconnected world, where individuals and organizations communicate across vast geographical distances via the internet, ensuring the privacy and security of data is paramount. Traditional symmetric or private-key cryptography, which relies on a single shared secret key between sender and receiver, faces a significant challenge: how to securely exchange this secret key in the first place. Imagine trying to send sensitive information to a friend thousands of kilometers away; while symmetric algorithms can encrypt the data, the initial key exchange itself becomes a vulnerability. Sharing keys over conventional channels like telephone calls is insecure, as these mediums can be intercepted. This fundamental problem highlights the need for a more robust and scalable solution for secure key distribution and communication in a distributed environment like the internet.

Public-key cryptography emerges as a revolutionary paradigm to address these challenges. Unlike its symmetric counterpart, it eliminates the need for a pre-shared secret key, offering a secure mechanism for both encryption and authentication. This lecture introduces the foundational concepts of public-key cryptography, delving into its operational principles, key generation, and practical applications. We will explore one of the most widely adopted public-key algorithms, RSA, understanding its mathematical underpinnings and its critical role in securing modern internet communications, while also considering its future vulnerabilities.

Core Concepts

Term	Definition	Significance
Public-Key Cryptography	A cryptographic system that uses a pair of mathematically linked keys: a public key, which can be widely distributed, and a private key, which is kept secret by its owner.	Solves the key distribution problem inherent in symmetric cryptography, enabling secure communication between parties without prior shared secrets.
Symmetric Key Cryptography	A cryptographic system where the same secret key is used for	Efficient for bulk data encryption but faces

Term	Definition	Significance
	both encryption and decryption.	significant challenges in securely distributing the shared secret key, especially over insecure channels.
Public Key	One half of a key pair in public-key cryptography, freely distributed and known to anyone who wishes to communicate securely with the owner.	Used by others to encrypt messages for the owner or to verify digital signatures created by the owner.
Private Key	The other half of a key pair in public-key cryptography, kept secret and known only to its owner.	Used by the owner to decrypt messages encrypted with their public key or to create digital signatures.
RSA Algorithm	A widely used public-key cryptosystem named after its inventors (Rivest, Shamir, Adleman), based on the computational difficulty of factoring large integers.	Forms the backbone of much of modern internet security, used for secure data transmission, digital signatures, and key exchange.
Euler Totient Function ($\phi(n)$)	A mathematical function that counts the number of positive integers up to a given integer n that are relatively prime to n . For $n = p * q$ (where p and q are prime), $\phi(n) = (p-1)(q-1)$.	A crucial component in the RSA key generation process, determining the modular arithmetic properties required for encryption and decryption.
Digital Signature	An electronic signature that uses public-key cryptography to verify the authenticity and integrity of a digital message or document.	Provides assurance to the recipient that the message originated from the claimed sender and has not been altered in transit.

Term	Definition	Significance
Key Exchange	The process by which two parties establish a shared secret key over an insecure communication channel.	Public-key cryptography, particularly RSA, is often used to securely exchange a symmetric key, which is then used for faster bulk data encryption.
Quantum Computing	A new type of computing that uses quantum-mechanical phenomena like superposition and entanglement to perform computations.	Poses a significant future threat to current public-key cryptosystems like RSA, as quantum algorithms can efficiently solve the underlying mathematical problems.
Post-Quantum Cryptography	Cryptographic algorithms designed to be secure against attacks by quantum computers.	An active area of research and development, aiming to replace current vulnerable cryptosystems before large-scale quantum computers become available.

Detailed Analysis

1. Public-Key Cryptography: A Paradigm Shift in Key Management

Public-key cryptography, also known as asymmetric cryptography, fundamentally differs from symmetric key systems by employing a pair of keys for each participant: a public key and a private key. This innovation addresses the critical key distribution problem faced by symmetric systems, especially in large, distributed networks like the internet.

- Key Management in Symmetric Systems:** In symmetric cryptography, a single secret key is shared between two communicating parties. If 'n' parties need to communicate securely with each other, the number of shared keys required grows quadratically, approximately $n * (n-1) / 2$. This becomes unmanageable for a large number of users, as securely distributing and managing these many keys is a logistical nightmare.

- **Key Management in Public-Key Systems:** In contrast, each party in a public-key system generates its own unique pair of keys: one public and one private. The public key is openly distributed and can be known by anyone, while the private key is kept strictly secret by its owner. This means for 'n' parties, there are only $2 * n$ keys in total (n public keys and n private keys). This drastically simplifies key management.
- **Core Principle:** The mathematical relationship between the public and private keys is such that data encrypted with one key can only be decrypted with the other. This interchangeability is central to its functionality for both encryption and authentication.

2. Applications of Public-Key Cryptography: Encryption and Authentication

Public-key cryptography offers two primary applications: secure encryption and digital authentication.

- **Secure Encryption:**
 - When sender A wants to send a confidential message to receiver B, A uses B's publicly available public key to encrypt the plaintext.
 - The resulting ciphertext is then sent to B.
 - Only B, possessing the corresponding private key, can decrypt the ciphertext back into the original plaintext.
 - This ensures that even if the ciphertext is intercepted, no one other than B can read the message, as B's private key is unique and secret. This mechanism allows many parties to send encrypted messages to B, but only B can decode them.
- **Digital Authentication:**
 - For authentication, the roles of the keys are reversed. If sender A wants to prove its identity to receiver B, A encrypts a piece of data (e.g., a random text or a message digest) using its own private key.
 - A then sends this "signed" data to B.
 - B, upon receiving the data, uses A's publicly known public key to decrypt it.
 - If the decryption is successful and yields a recognizable or expected result, B can be confident that the message originated from A, because only A possesses the private key capable of creating data that can be successfully decrypted by A's public key.
 - This process does not guarantee secrecy (as anyone with A's public key can decrypt it), but it guarantees the authenticity of the sender.

- **Interchangeability of Keys:** A crucial property of public-key algorithms is that encryption and decryption are reversible. If you encrypt with the public key, you decrypt with the private key. Conversely, if you encrypt with the private key, you decrypt with the public key. The first mechanism is primarily used for confidentiality (encryption), while the second is used for authentication.

3. RSA Algorithm: Key Generation

The RSA algorithm, named after Rivest, Shamir, and Adleman, is a cornerstone of public-key cryptography. Its security relies on the computational difficulty of factoring large prime numbers. The key generation process involves several steps:

- **Step 1: Select Two Large Prime Numbers (p and q):**
 - The first step is to choose two distinct, very large prime numbers, p and q . These numbers are not small (e.g., 5 and 13), but typically hundreds of digits long (e.g., 300 digits or more), corresponding to 1024 or 2048 bits. The larger these primes, the more secure the system.
- **Step 2: Calculate n :**
 - Compute n as the product of p and q : $n = p * q$. This n will be part of both the public and private keys.
- **Step 3: Calculate Euler Totient $\phi(n)$:**
 - Compute Euler's totient function for n , denoted as $\phi(n)$. For two prime numbers p and q , $\phi(n) = (p-1) * (q-1)$. This value represents the number of positive integers less than n that are relatively prime to n .
- **Step 4: Select Public Exponent e :**
 - Choose an integer e such that $1 < e < \phi(n)$ and e is relatively prime to $\phi(n)$. This means the greatest common divisor (GCD) of e and $\phi(n)$ must be 1. Common choices for e include 3, 17, or 65537.
- **Step 5: Determine Private Exponent d :**
 - Calculate d such that $d * e \equiv 1 \pmod{\phi(n)}$. This means d is the multiplicative inverse of e modulo $\phi(n)$. This d can be found using the extended Euclidean algorithm.
- **Step 6: Form Key Pairs:**
 - The **public key** consists of the pair (e, n) . This key is made public.
 - The **private key** consists of the pair (d, n) . This key must be kept secret.

4. RSA Algorithm: Encryption and Decryption

Once the key pairs are generated, RSA encryption and decryption are straightforward mathematical operations.

- **Encryption:**

- To encrypt a plaintext message M (which is treated as an integer, where $0 \leq M < n$), the sender computes the ciphertext C using the recipient's public key (e, n) :

$$C = M^e \bmod n$$

- The message M is typically a stream of bits interpreted as a large integer. The only constraint is that M must be less than n .
- This operation involves raising a potentially very large number M to the power of e and then taking the modulo n . While conceptually simple, this can be computationally intensive for large numbers, making RSA slower than symmetric algorithms.

- **Decryption:**

- To decrypt the ciphertext C back to the original plaintext M , the recipient uses their private key (d, n) :

$$M = C^d \bmod n$$

- Substituting $C = M^e$, we get $M = (M^e)^d \bmod n = M^{(e*d)} \bmod n$.
- Since $e*d \equiv 1 \pmod{\phi(n)}$, it can be proven that $M^{(e*d)} \equiv M \pmod{n}$, thus recovering the original message M .
- Similar to encryption, this operation involves raising a large number to a large power modulo n , requiring efficient modular exponentiation algorithms.

- **Illustrative Example:**

- Let $p = 7$, $q = 17$.
- $n = p * q = 7 * 17 = 119$.
- $\phi(n) = (p-1)(q-1) = (7-1)(17-1) = 6 * 16 = 96$.
- Choose $e = 5$ (since $\text{GCD}(5, 96) = 1$ and $5 < 96$).
- Calculate d such that $5 * d \equiv 1 \pmod{96}$. The smallest d is 77 (since $5 * 77 = 385$, and $385 \bmod 96 = 1$).
- Public Key: $(5, 119)$. Private Key: $(77, 119)$.

- **Encryption (Plaintext $M = 19$):**

$$C = 19^5 \bmod 119 = 2476099 \bmod 119 = 66.$$

- **Decryption (Ciphertext C = 66):**

$M = 66^{77} \bmod 119 = 19$. (This calculation is complex but yields 19).

5. Security and Hybrid Applications of RSA

The security of RSA hinges on the computational difficulty of factoring large integers. If an adversary can factor n into its prime components p and q , they can then easily calculate $\phi(n)$ and subsequently d from e , thereby breaking the private key.

- **Factoring Problem:** Currently, there is no known efficient classical algorithm to factor the product of two very large prime numbers. This makes it computationally infeasible for an attacker to derive the private key from the public key (e, n) . The large size of p and q (e.g., 1024-bit or 2048-bit numbers) ensures this difficulty.
- **Performance Considerations:** While highly secure, RSA is significantly slower than symmetric key algorithms like DES or AES due to the complex modular exponentiation operations involving very large numbers.
- **Hybrid Cryptosystems:** To leverage RSA's secure key exchange capabilities while maintaining high performance for bulk data encryption, a hybrid approach is commonly used:
- **Key Exchange:** A sender generates a random, one-time symmetric key (e.g., an AES key) for a specific communication session. This symmetric key is then encrypted using the recipient's RSA public key.
- **Secure Transmission:** The encrypted symmetric key is sent to the recipient. Since the symmetric key is relatively small, RSA encryption is acceptably fast for this step.
- **Decryption and Bulk Encryption:** The recipient uses their RSA private key to decrypt the symmetric key. Both parties now securely share the same symmetric key.
- **Data Encryption:** All subsequent large data transfers for that session are encrypted and decrypted using the much faster symmetric key algorithm (e.g., AES) with the newly exchanged key.
- This hybrid model is the foundation of secure communication protocols like HTTPS, used in internet banking and other online transactions.

6. The Quantum Threat and Post-Quantum Cryptography

Despite its current robustness, RSA faces a significant future threat from quantum computing.

- **Quantum Computing's Impact:** Quantum computers, which leverage principles of quantum mechanics, are theoretically capable of running algorithms (like Shor's algorithm) that can efficiently factor large numbers. If a sufficiently powerful quantum computer is built, it could break RSA by factoring n into p and q in a feasible amount of time.
- **Current Status:** While small-scale quantum computers exist, a "cryptographically relevant" quantum computer capable of breaking RSA is still years away. However, research and development in quantum computing are progressing rapidly.
- **Post-Quantum Cryptography (PQC):** Recognizing this impending threat, cryptographers are actively researching and developing "post-quantum cryptographic algorithms." These are new cryptographic schemes designed to be resistant to attacks from quantum computers, ensuring the continued security of digital communications in a post-quantum era. Companies and organizations are already beginning to upgrade their infrastructure to incorporate PQC methods.

Key Takeaways

- [] Public-key cryptography uses a pair of keys (public and private) per user, solving the key distribution problem of symmetric systems.
- [] The public key is shared, while the private key is kept secret.
- [] For encryption, the sender uses the recipient's public key, and the recipient decrypts with their private key.
- [] For authentication, the sender encrypts with their private key, and the recipient verifies with the sender's public key.
- [] RSA key generation involves selecting large prime numbers p and q , calculating $n = p \cdot q$ and $\phi(n) = (p-1)(q-1)$, then finding e and d based on $\phi(n)$.
- [] RSA encryption is $C = M^e \bmod n$, and decryption is $M = C^d \bmod n$.
- [] The security of RSA relies on the computational difficulty of factoring large numbers into their prime factors.
- [] RSA is slower than symmetric algorithms, so it's often used in hybrid systems for secure key exchange, with symmetric algorithms handling bulk data encryption.

- [] Quantum computers pose a future threat to RSA by potentially being able to efficiently factor large numbers, necessitating the development of post-quantum cryptography.
- [] Modern internet security (e.g., HTTPS) heavily relies on this hybrid approach, using RSA for initial key exchange.

Conclusion

Public-key cryptography, particularly the RSA algorithm, has fundamentally transformed the landscape of secure digital communication. By introducing the concept of distinct public and private keys, it elegantly solved the long-standing problem of secure key distribution, enabling confidential and authenticated exchanges over insecure networks like the internet. Its applications span from encrypting sensitive data and verifying digital identities through digital signatures to facilitating secure key exchanges for faster symmetric encryption, forming the bedrock of modern online transactions and secure web browsing.

However, the cryptographic world is dynamic, constantly evolving to counter new threats. The advent of quantum computing presents a significant challenge to RSA's foundational security, as future quantum machines could potentially break its underlying mathematical problem. This looming threat underscores the ongoing importance of cryptographic research, driving the development of post-quantum cryptographic algorithms. While RSA remains robust for now, its eventual replacement highlights the continuous race between cryptographers and adversaries, ensuring that the principles of secure communication remain at the forefront of technological advancement.